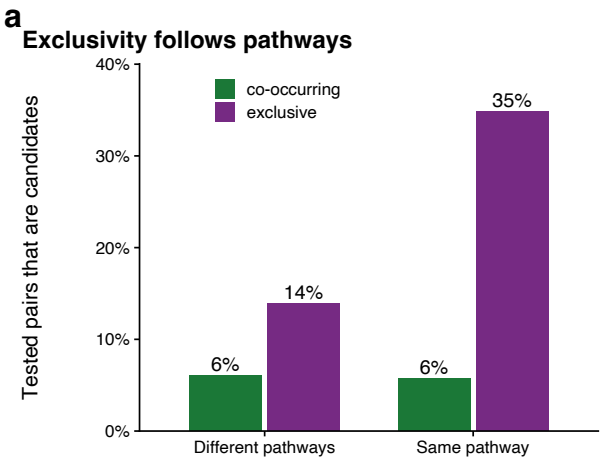
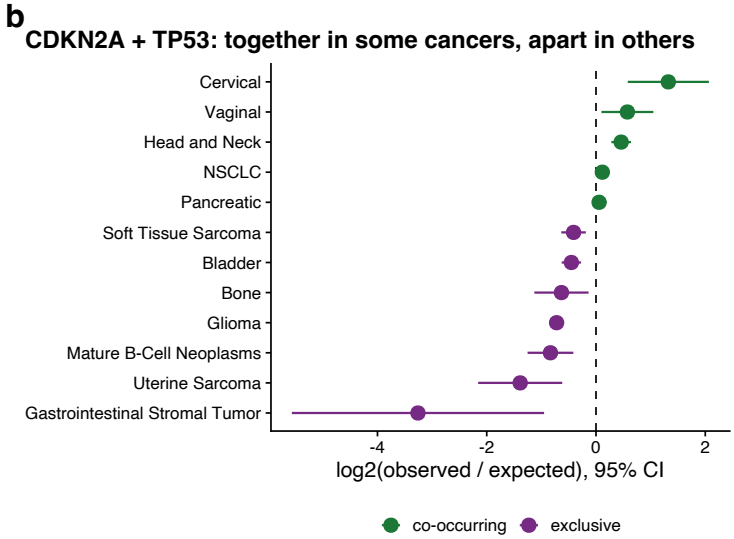
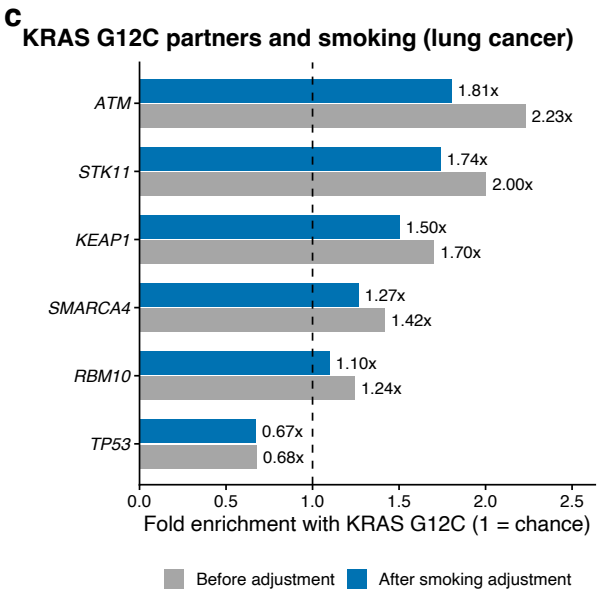
# Phase 4: exclusivity follows pathways, and pairs change with the cancer type



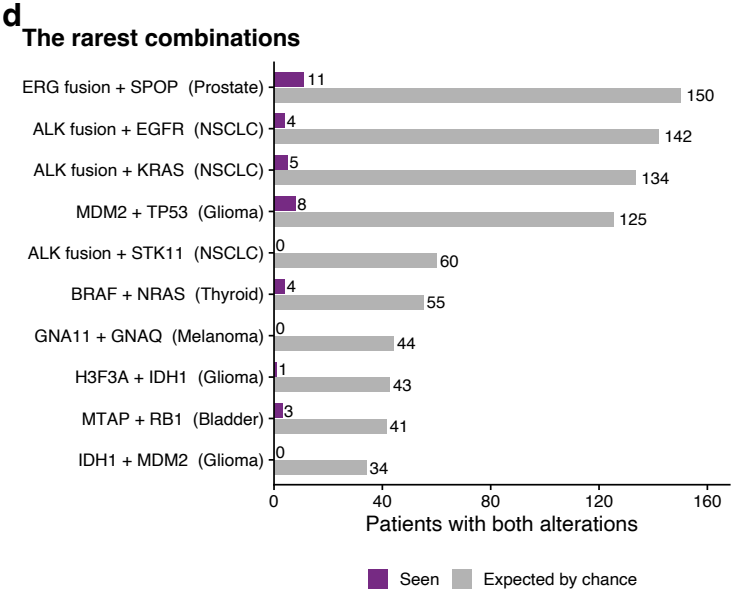

Genes in the same pathway avoid each other 2.5x more often: one hit per pathway is enough.




Cancer types where the pair is significant ($q < 0.05$).




Smoking explains part of G12C + STK11/KEAP1 and all of G12C + RBM10.




Combinations seen far less often than chance predicts (expected >= 20 patients).